

EduPilot AI: Adaptive Learning Platform with AI-Powered Learning Path Optimization

Review-1 Project Proposal

Theme: AI/Technology-Enabled Innovation for Education and Learning Ecosystems

1. Problem Statement

Students often receive identical learning material regardless of their prior knowledge. Existing AI assistants answer questions but rarely diagnose prerequisite gaps or generate personalized learning paths aligned with institutional course material.

2. Motivation & Target Users

Target users are university students and faculty. The platform aims to improve learning efficiency, provide adaptive explanations, identify weak concepts, and help teachers monitor class-wide learning progress.

3. Existing Solutions & Gap

Existing solutions include ChatGPT, Gemini, Moodle, Google Classroom and Khanmigo. While these provide tutoring or LMS capabilities, they do not combine teacher-controlled content, prerequisite-aware learning paths, student knowledge modelling, and analytics in a single adaptive platform. EduPilot AI fills this gap using Retrieval-Augmented Generation (RAG), knowledge graphs, and adaptive recommendation.

4. Proposed Idea

Teachers upload notes, PDFs and assignments. The system extracts concepts, builds a prerequisite knowledge graph, indexes documents for retrieval, and creates quizzes. Students interact through an AI tutor that answers only from uploaded content, estimates concept mastery, recommends the next topics to study, and adapts explanations to the student's level.

5. Core Modules

Teacher Dashboard; Student Dashboard; Document Processing; Vector Database (RAG); Knowledge Graph; Learning Path Optimizer; Adaptive AI Tutor; Progress Analytics.

6. AI Techniques & Syllabus Mapping

Information Retrieval (RAG), Information Extraction (concept extraction), Intelligent Agent (adaptive tutor), Search & Graph algorithms (knowledge graph traversal), Planning (learning roadmap), NLP (question answering), Knowledge Representation (concept graph).

7. Objectives

1. Build an adaptive learning platform.
2. Generate personalized learning paths.
3. Provide syllabus-grounded AI responses.
4. Detect knowledge gaps.
5. Generate quizzes and flashcards.
6. Provide teacher analytics.

8. Expected Output

Working web prototype, AI tutor, teacher dashboard, student dashboard, knowledge graph visualization, personalized learning path generator, analytics dashboard, evaluation results.

9. Technology Stack

Frontend: Next.js + Tailwind CSS

Backend: FastAPI

Database: PostgreSQL

AI: Python, Hugging Face Transformers, Sentence Transformers, ChromaDB/FAISS, NetworkX

LLM: Local open-source model (Phi-3/Gemma/Qwen).

10. Two-Month Development Plan

Week 1: Setup & Auth

Week 2: Teacher uploads

Week 3: RAG pipeline

Week 4: Knowledge graph

Week 5: Learning path optimizer

Week 6: Adaptive tutor

Week 7: Analytics

Week 8: Testing & deployment.

11. Novelty

Unlike a generic chatbot, EduPilot AI models student knowledge, respects prerequisite relationships, personalizes explanations, and generates algorithmic learning paths while remaining grounded in teacher-provided material.